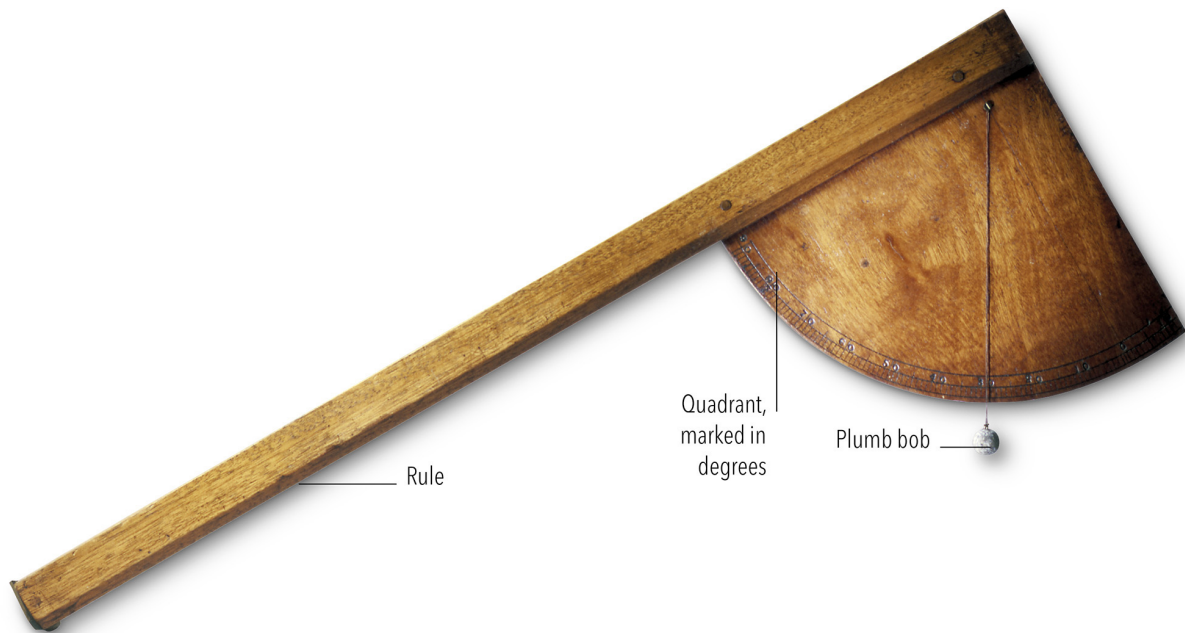


densely packed mine that triggered the Battle of the Crater at Petersburg. Finally, Grant won the artillery duel with only his field artillery and some heavy ordnance loaned to him by Union gunboats.

80,000 Estimated number of mortar shells fired during the ten-month siege of Petersburg.



Wooden quadrant

This simple quadrant was used by gunners to check the elevation of a cannon in order to give the shell the correct angle of trajectory to hit its target.

Heavy artillery old and new

Siege artillery was often dwarfed by coastal artillery: large-caliber Columbiads and similar monsters fired from fixed platforms. By the time of the Civil War, however, these great smoothbores and their enclosing forts were becoming obsolete, as new artillery with rifled—or grooved—barrels came into use. These weapons were faster to fire, more accurate, and had a greater range. The thick masonry walls of Fort Pulaski outside Savannah were reduced to rubble in two days in April 1862 by the new rifled ordnance of the Union fleet. Earth forts fared better, however, as they absorbed the blow and shock of an exploding shell. Battery Wagner outside Charleston,